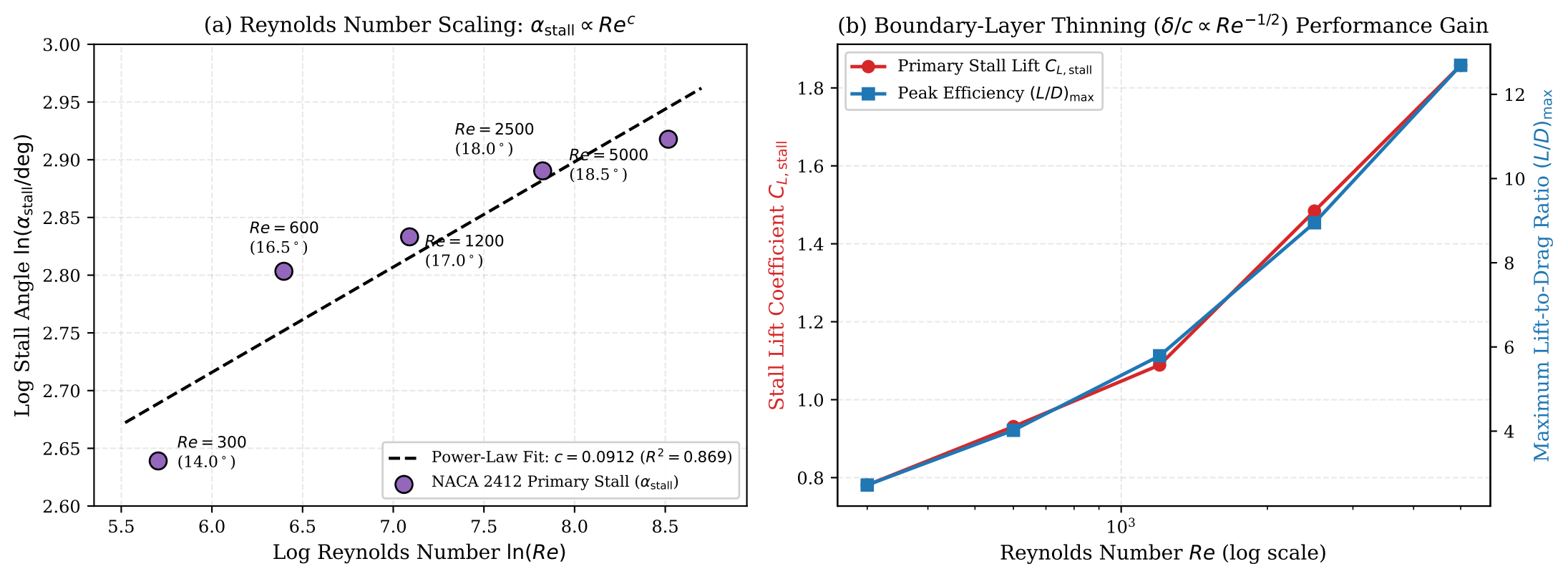
(a) Reynolds Number Scaling: $\alpha_{stall} \propto Re^c$
Log Stall Angle $\ln(\alpha_{stall}/\deg)$
Log Reynolds Number $\ln(Re)$
$Re = 300$ (14.0°)
$Re = 600$ (16.5°)
$Re = 1200$ (17.0°)
$Re = 2500$ (18.0°)
$Re = 5000$ (18.5°)
Power-Law Fit: $c = 0.0912$ ($R^2 = 0.869$)
NACA 2412 Primary Stall ($\alpha_{stall}$)
(b) Boundary-Layer Thinning ($\delta/c \propto Re^{-1/2}$) Performance Gain
Primary Stall Lift $C_{L,stall}$
Peak Efficiency $(L/D)_{max}$
Stall Lift Coefficient $C_{L,stall}$
Maximum Lift-to-Drag Ratio $(L/D)_{max}$
Reynolds Number $Re$ (log scale)